

Rishabh Gosain

AI Engineer | LLMs, Machine Learning & Python
+1 (934) 246-4618 | rgosain2001@gmail.com | [LinkedIn](#) | [Portfolio](#) | [GitHub](#)

SUMMARY

AI Engineer with hands-on experience building production-grade LLM systems, multi-agent architectures, and RAG pipelines using Python, LangChain, HuggingFace, Weaviate, and FastAPI. Skilled in fine-tuning open-source models (QLoRA/PEFT), designing LLM evaluation frameworks, and deploying end-to-end AI workflows from data ingestion and vector retrieval to containerized cloud deployment.

WORK EXPERIENCE

Stony Brook University
Research Assistant

Stony Brook, New York
Aug. 2025 - May. 2026

- Fine-tuned Mistral-7B on domain-specific Q&A data using QLoRA (4-bit quantization via PEFT/HuggingFace) on GPU — reducing hallucination rate by 34% over base model and achieving 2.1x improvement in domain-specific ROUGE-L score, demonstrating hands-on model adaptation beyond API-level LLM usage.
- Built a systematic LLM evaluation framework benchmarking fine-tuned model vs. base LLM vs. RAG pipeline across 100+ test queries — tracking ROUGE, BERTScore, latency, and hallucination metrics with experiment logging via W&B, reflecting production-grade model evaluation and MLOps practices.

WTC 911 Stony Brook Hospitals
Senior Informatics Intern

Commack, New York
May. 2025 - Aug. 2025

- Built automated ETL pipelines using Python and SQL to ingest, transform, validate, and process healthcare datasets, reducing manual operational effort by 90%.
- Developed scalable data preprocessing workflows for large healthcare datasets, ensuring high-quality inputs for analytics and machine learning.
- Collaborated with cross-functional teams to implement data quality frameworks, schema validation rules, and automated data processing workflows.

Coding Blocks Pvt. Ltd.
AI Engineer

Delhi, India
Jun. 2023 - Jul. 2024

- Built an NLP-powered text analysis system using HuggingFace Transformers for sentiment classification and topic modeling on large-scale user-generated text data, automating insight extraction and reducing manual review effort by 40%.
- Designed and deployed Python-based ML pipelines and recommendation workflows to process behavioral data at scale, improving user engagement rates by 25% and delivering automated insights via internal dashboards to support data-driven decision-making.

PROJECTS

AI-Powered Financial Intelligence Platform

- Built a production-grade RAG pipeline using Python, LangChain, FAISS, and LLM APIs — ingesting and indexing structured knowledge sources, engineering retrieval workflows for context-aware reasoning, and generating automated analytical reports from large-scale structured datasets.
- Designed LangChain-based agentic reasoning workflows integrating external APIs and vector retrieval to deliver real-time, contextually grounded insights from fragmented data sources — demonstrating end-to-end ownership from data ingestion and context engineering to automated report generation.

Enterprise AI Agent Platform with RAG & Multi-Agent Orchestration

- Built a production-grade multi-agent AI platform using Python, LangChain, FastAPI, and MCP — implementing a coordinator-agent architecture with specialized sub-agents for RAG-based document retrieval (Weaviate), predictive risk scoring (XGBoost), real-time web research (Tavily), and natural language analytics, containerized via Docker and deployed on cloud infrastructure.
- Engineered a Weaviate-backed RAG pipeline with chunk-level embedding, semantic retrieval, and LLM reasoning — benchmarking RAG vs. baseline LLM responses across 50+ test queries, optimizing prompt engineering and retrieval strategies to reduce hallucination on domain-specific content.
- Integrated an XGBoost predictive scoring engine and LangChain natural language query layer into a Streamlit dashboard with experiment tracking via W&B — enabling non-technical users to interrogate structured enterprise datasets in plain English and receive automated insight reports.

SKILLS

Programming Languages	: Python, SQL, JavaScript, R
AI & Machine Learning	: Machine Learning, Deep Learning, Natural Language Processing (NLP), Large Language Models (LLMs), Generative AI, Computer Vision, Scikit-learn, PyTorch, Model Evaluation, Feature Engineering, Time Series Forecasting
Generative AI & LLMs	: LangChain, Prompt Engineering, Retrieval-Augmented Generation (RAG), AI Workflow Automation, LLM Applications, API Integration, Weaviate, FAISS, Fine-Tuning, QLoRA, PEFT, Multi-Agent Systems, MCP
Data Engineering	: ETL Pipelines, Data Processing, Data Validation, Data Modeling, Data Warehousing, Data Integration, Data Transformation, Workflow Automation
Data Analysis	: Pandas, NumPy, Statistical Analysis, Data Cleaning, Exploratory Data Analysis (EDA)
Databases & Cloud	: MySQL, PostgreSQL, SQL Server, MongoDB, Snowflake, BigQuery, AWS, Azure
Visualization & Reporting	: Tableau, Power BI, Jupyter Notebook, Data Visualization, Dashboard Development
Tools & Technologies	: Git, Docker, VS Code, CI/CD, REST APIs

EDUCATION

Stony Brook University, Stony Brook, NY
Master of Science in Data Science

Aug. 2024 - May 2026

Guru Gobind Singh Indraprastha University, Delhi, India
Bachelor of Technology in Computer Science

Aug. 2020 - Jul. 2024